DE LA SALLE UNIVERSITY-DASMARINAS
COLLEGE OF INFORMATION AND COMPUTER STUDIES
INFORMATION TECHNOLOGY DEPARTMENT

COURSE SYLLABUS
First Semester, SY 2024-2025

COURSE CODE	**:**	S-ITPE 001LA
COURSE TITLE	**:**	IT Professional Elective
COURSE TYPE	**:**	Laboratory
COURSE CREDIT	**:**	One Unit
LEARNING MODALITY	**:**	Fully Onsite
PRE-REQUISITES	**:**	None
CO-REQUISITES	**:**	S-ITPE 001
PROFESSOR	**:**	Ms. Azenith R. Mojica
CONSULTATION HOURS	**:**	Thursday: 8:00-10:00 ONLINE Friday: 7:00-12:00 COS 100-C
CONTACT INFORMATON	**:**	armojica@dlsud.edu.ph (046) 4811900 local 3134

VISION

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:

This course covers the fundamentals of object-oriented programming and applications using Python programming language. Covers a range of topics in object-oriented including information hiding, polymorphism, method overriding, method overloading, class inheritance, static and dynamic binding, and generic programming. Emphasis is also placed on object-oriented design approaches. This course provides hands-on experience using Python programming language through both individual assignment and group programming projects.

COURSE LEARNING OUTCOMES:

By the end of this course, students can:

CLO 1. explore Python programming language Integrated Development Environment (IDE);

CLO 2. recall the basic concepts of programming.

CLO 3. examine program functionality.

CLO 4. explain the fundamental concepts of object-oriented programming.

CLO 5. design programs using object-oriented programming concepts; and

CLO 6. demonstrate 21st century skills in all learning activities.

LEARNING PLAN:

Learning Outcomes for Midterm Period

Course Learning Outcomes	Topic Learning Outcomes
CLO1. Explore Python programming language Integrated Development Environment (IDE).	TLO1. Install and explore the features of using an IDE. TLO2. Explore available online Python compilers.
CLO2. Recall the basic concepts of programming.	TLO3. Apply basic I/O Statements. TLO4. Review the basic control structures.
CLO 3. Examine program functionality.	TLO5. Evaluate program execution by tracing program output. TLO6. Test correctness of program. TLO7. Validate program input.

Module No and Learning Outcomes	Week	Teaching-Learning Activities / Assessment Strategies	Technology Enabler	Allotted Hours
Module 0	Week 1 Aug 19-24	• Gospel Reading, Reflection, and Prayer • Discussion of Course Syllabus **Module 0: Course Introduction** • Enabling Assessment (Research on the History of Python Programming Language and different Python IDEs)	Schoolbook PowerPoint	3
Module 1 CLO1 TLO1 TLO2	Week 2 Aug 26-31	• Gospel Reading, Reflection, and Prayer • Discussion of **Module 1: Introduction to Python Programming** • Enabling Assessment (A 20-item Laboratory Activity about the Basics of Python Programming)	Schoolbook PowerPoint Python IDE	3
Module 1 CLO1 TLO1 TLO2	Week 3 Sept 2-7	• Gospel Reading, Reflection, and Prayer • Discussion of **Module 2: Decision-making Statement** • Enabling Assessment (A 20-item Laboratory Activity about Python's Decision -making Statements)	Schoolbook PowerPoint Python IDE	3

Module 2 CLO 2 TLO3 TLO4	Week 4 Sept 9-14	• Gospel Reading, Reflection, and Prayer • Enabling Assessment (A 20-item Laboratory Activity about Python's Decision -making Statements)	Schoolbook PowerPoint Python IDE	3
LONG EXAM	Week 5 Sept 16-21	• Gospel Reading, Reflection, and Prayer • Discussion of **Module 3: Looping Statements** • Enabling Assessment (A 20-item Laboratory Activity about Python's Looping Statements)	Schoolbook PowerPoint Python IDE	3
Module 3 CLO 3 TLO5 TLO6 TLO7	Week 6 Sept 23-28	• Gospel Reading, Reflection, and Prayer • Enabling Assessment (A 20-item Laboratory Activity about Python's Decision -making Statements)	Schoolbook PowerPoint Python IDE	3
	Week 7 Sept 30 - Oct 5	• Gospel Reading, Reflection, and Prayer • Completion Week • Consultation Week		3
MIDTERM EXAM WEEK	Week 8 Oct 7-12	• Gospel Reading, Reflection, and Prayer • Midterm Summative Assessment on Modules 1-3		3
TOTAL				24

Learning Outcomes for Final Period

Course Learning Outcomes	Topic Learning Outcomes
CLO4. Explain the fundamental concepts of object-oriented programming.	TLO8. Discuss the characteristics of Object-Oriented Programming Language. TLO9. Differentiate class and objects. TLO10. Differentiate access levels of class members.
CLO5. Design programs using object-oriented programming concepts.	. TLO11. Explain the importance of inheriting classes. TLO12. Devise a class that inherits from created user-defined classes. TLO13. Explains how polymorphism works. TLO14. Handle program exceptions.
CLO6.Demonstrate 21st century skills in all learning activities.	*Collaboration, Skilled Communication, Knowledge Construction, Self- Regulation, Real-world Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)*

Module No and Learning Outcomes	**Week**	**Teaching-Learning Activities / Assessment Strategies**	**Technology Enabler**	**Allotted Hours**
Module 4 CLO4 TLO8 TLO9 TLO10	Week 10 Oct 21-26	• Gospel Reading, Reflection, and Prayer • Distribution of Midterm Grades • Enabling Assessment (Research work on User-defined Methods in Python- by pair)	Schoolbook PowerPoint Python IDE	3

	Week 11 Oct 28 - Nov 2	***Independent Learning Week***		
Module 4 CLO4 TLO8 TLO9 TLO10	Week 12 Nov 4-9	• Gospel Reading, Reflection, and Prayer • Discussion of **Module 4: User-defined Methods** • Enabling Assessment (A 20-item Laboratory Activity about User-defined Methods)	Schoolbook PowerPoint Python IDE	3
Module 4 CLO4 TLO8 TLO9 TLO10	Week 13 Nov 11-16	• Gospel Reading, Reflection, and Prayer • Enabling Assessment (A 20-item Laboratory Activity about User-defined Methods)	Schoolbook Python IDE	3
Module 5 CLO5 TLO11 TLO12	Week 14 Nov18-23	• Gospel Reading, Reflection, and Prayer • Discussion of **Module 5: Classes and Objects** • Enabling Assessment (A 20-item Laboratory Activity on Classes and Objects)	Schoolbook PowerPoint Python IDE	3

Module 6 CLO5 CLO6 TLO13 TLO14	Week 15 Nov 25-30	• Gospel Reading, Reflection, and Prayer • Discussion of **Module 6: Encapsulation, Inheritance, Abstraction and Polymorphism** • Enabling Assessment (A 20-item Laboratory Activity on encapsulation, Inheritance, Abstraction and Polymorphism)	Schoolbook PowerPoint Python IDE	3
	Week 16 Dec 2-7	• Gospel Reading, Reflection, and Prayer • Completion Week • Consultation Week		3
	Week 17 Dec 9-14	• Gospel Reading, Reflection, and Prayer • Final Term Summative Assessment on Modules 4-6		3
TOTAL				24

GRADING SYSTEM:

Midterm	**%**	**Final**	**%**
Enabling Assessments	40%	Enabling Assessments	40%
Summative Assessment (Midterm Examination)	60%	Summative Assessment (Final Examination)	60%
	100%		100%

FINAL GRADE = $\frac{\text{Midterm Grade + Final Grade}}{2}$

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Onsite Learning Mode. All activities will be done in the laboratory classroom and field.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar.

3. Laboratory Policies
   a. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
   b. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless a special permission is sought. Cellular phones also cannot be used as calculator during examination.
   d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

4. Communication and Feedback
   a. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
      i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
      ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through Schoolbook messaging feature.
      iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
   b. No use of foul, curse, and other bad words, whether in written or verbal. No shouting. No flooding of chat and messages. No selling.

5. Attendance Policy
   a. Attendance Policy for Onsite Classes is scheduled based on the learning plan.
   b. The onsite meeting will only be held at the schedule indicated in the registration form.
   c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

6. Intellectual Property Policy
   a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
   b. Using any illegally obtained software is strictly prohibited.
   c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
   d. Giving of password to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 account other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

7. Other Policies
   a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.

REFERENCES:

On-Site References

Provider	Reference Material
AEA Library QA 76.73.P98.L510 2019	Lee, Wei-Meng. 2019. Python Machine. (Indianapolis, IN): John Wiley and Sons
AEA Library QA 76.73.P98 .M378 2022	Mastrodomenico, R. 2022. The Python Book. (Hoboken, NJ): Wiley

Online References

Call Number/ E-provider	Reference Material
E-Book Accession Number: 1991381	Bhasin, H. 2019. Python Basics: A Self-Teaching Introduction. (Dulles, Virginia): Mercury Learning & Information.
E-Book Accession Number: 2291496	Bird, A. et.al. 2019. The Python Workshop: Learn to Code in Python and Kickstart Your Career in Software Development or Data Science. (Birmingham): Packt Publishing.
Web Resource	Amos, David, 2023.Object-oriented Programming (OOP) in Python 3 [accessed 2024 July 30] https://realpython.com/python3-object-oriented-programming/

Prepared: **Information Technology Department**
1st Semester 2024-2025

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